

# Influence of understory vegetation on fungal communities and 2,4,6-trichloroanisole contamination in raw cork from standing trees

Salvatore Seddaiu<sup>a,1,\*</sup>, Camilla Morittu<sup>a,c,2</sup>, Luca Sarais<sup>a,3</sup>, Antonio Mulas<sup>a,4</sup>,  
Adriana Vilia<sup>a</sup>, Gianfranco Marzeddu<sup>a</sup>, Giuseppino Pira<sup>a</sup>, Pino A. Ruiu<sup>a</sup>, Edoardo Scali<sup>b,5</sup>,  
Giovanni Piras<sup>a</sup>, Francesca Meloni<sup>a</sup>, Marco Serra<sup>a</sup>, Margherita Addis<sup>a</sup>, Marco Caredda<sup>a,6</sup>,  
Stefania Fanni<sup>a</sup>, Vittorio Alessandro Marrone<sup>a</sup>, Matteo Garbelotto<sup>b,7</sup>, Bruno Scanu<sup>c,8</sup>,  
Franco Pampiro<sup>a</sup>

<sup>a</sup> Agricultural Research Agency of Sardinia – AGRIS Sardegna, Sassari, Italy

<sup>b</sup> Department of Environmental Science, Policy and Management (ESPM), University of California, Berkeley, CA, USA

<sup>c</sup> Department of Agricultural Sciences, University of Sassari, Viale Italia 39/A, Sassari, Italy

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## ABSTRACT

Cork oak forests are of considerable economic importance, they serve as pastures, and they produce the raw material used by the cork industry. In many instances, the economic value of cork oak forests is closely linked to the demand for cork stoppers used in winemaking, and any decline in cork quality can have significant undesirable consequences. This is particularly relevant for cork taint associated with off-flavours caused by 2,4,6-trichloroanisole (TCA) and formed by microbial methylation of 2,4,6-trichlorophenol (TCP), a precursor that can originate either from anthropogenic sources, such as pollution or pesticides, or from natural cork components. This study compared fungal communities and TCA contamination levels in cork from forests with and without shrub understory, selected to assess the potential influence of understory vegetation on fungal composition and TCA formation. The correlation analysis showed a highly significant relationship ( $p = 0.005$ ) between understory presence and TCA levels, with a moderate positive correlation ( $r = 0.26$ ), suggesting that while understory presence contributes only partially to this correlation, it may also influence consequent environmental variables that contribute to TCA variability. These findings suggest that forest management practices aimed at controlling understory vegetation are crucial for reducing cork taint contamination and enhancing cork quality. At the same time, maintaining fungal and plant diversity within cork oak stands requires management approaches that preserve ecological integrity while supporting the economic sustainability of cork production.

## 1. Introduction

Cork oak (*Quercus suber* L.) forests are among the most significant woodland ecosystems in the Mediterranean region due to their significant ecological and socio-economic value. These forests cover over 1.5

million hectares in European countries such as Portugal, Spain, France, and Italy, and an additional 0.7 million hectares in the Maghreb countries, including Morocco, Algeria, and Tunisia (Durrant et al., 2016; Matías et al., 2019). In Italy, cork oak forests cover approximately 168,000 ha, with Sardinia being the main cork-producing region (La Riccia

\* Corresponding author.

E-mail address: [saseddaiu@agrisricerca.it](mailto:saseddaiu@agrisricerca.it) (S. Seddaiu).

<sup>1</sup> <https://orcid.org/0000-0003-2364-4311>

<sup>2</sup> <https://orcid.org/0009-0006-0878-5246>

<sup>3</sup> <https://orcid.org/0009-0001-5385-0709>

<sup>4</sup> <https://orcid.org/0009-0001-5979-7769>

<sup>5</sup> <https://orcid.org/0000-0002-6112-2138>

<sup>6</sup> <https://orcid.org/0000-0002-5297-5702>

<sup>7</sup> <https://orcid.org/0000-0001-9737-0128>

<sup>8</sup> <https://orcid.org/0000-0002-0690-580X>

et al., 2023). Management heavily impacts the sustainability and productivity of cork oak forests. Management prescriptions dictate the frequency of cork harvesting but also regulate the presence of other tree species or shrubs, thus affecting the structure of these ecosystems (Bugalho et al., 2011). There are two main types of cork oak forest: forests with a high tree density and rich understory and open woodlands with sparse trees where the understory is cleared for pasture. The latter include the *montados* in Portugal, the *dehesas* in Spain, and the *pascoli arborati* in Sardinia (Sedda et al., 2011). Human activities have shaped these landscapes for centuries, with cork production emerging as the primary economic driver supporting a thriving global market due to cork's unique mechanical properties such as elasticity, impermeability, and insulation. A decline in the value of cork can lead to land abandonment, changes in land use, and increased human pressure on forests, possibly leading to the loss of cork oak woodlands (Bugalho et al., 2011; Azul et al., 2010).

Portugal is the leading world cork producer, placing approximately 100,000 tons of cork on the market annually, while Italy contributes around 6.161 tons, mainly from Sardinia (Oliveira and Pereira, 2020; Dettori and Filigheddu, 2017). Cork stoppers, particularly those used by the oenological industry, account for over 70 % of total cork production, highlighting the importance of cork quality throughout the production chain (Gil, 2014; Veloso et al., 2024). However, cork quality is critically affected by cork taint, primarily caused by the presence of 2,4,6-trichloroanisole (TCA) (Cravero, 2020; Cappellin et al., 2020). TCA contamination in cork stoppers is often associated with musty and mouldy odours in wine, leading to significant economic losses for the wine industry due to compromised product quality and consumer dissatisfaction (Cappellin et al., 2020; Tarasov et al., 2022). Contamination can occur at various stages, including during post-harvest processing or while the cork is still in the bark of standing trees (Tarasov et al., 2022). The formation of TCA results from the methylation of 2,4,6-trichlorophenol (TCP), which may originate from both anthropogenic and natural sources. Although chlorophenols were historically used as fungicides and wood preservatives, traces of these compounds and their methylated derivatives can also occur naturally in cork and other environmental matrices (Ruiz-Muñoz et al., 2025). More recently, TCA has been linked to the metabolic activity of fungal species inhabiting cork, highlighting a previously underestimated natural pathway (Tarasov et al., 2022; Goto et al., 2023; Veloso et al., 2024; Zhou et al., 2024; Ruiz-Muñoz et al., 2025). The methylation of TCP into TCA is mainly mediated by filamentous fungi, although bacteria can also contribute to the process. Current evidence suggests that TCP may reach cork tissue through gradual environmental adsorption, followed by microbial methylation. In addition, natural precursors of both TCP and TCA can originate within the cork bark itself, where phenolic compounds produced through lignin degradation and fungal metabolism may undergo chlorination to form TCP, subsequently methylated to TCA. (Álvarez-Rodríguez et al., 2002; Prak et al., 2007; Simpson and Sefton, 2007; Veloso et al., 2024; Zhou et al., 2024). Understanding the ecological dynamics of fungal communities associated with raw cork is crucial, as the presence of TCP alone does not lead to TCA formation without microbial intervention (Tarasov et al., 2022). Specific environmental conditions and the provenance of the cork can influence its chemical composition and the fungal communities, thereby affecting the likelihood of TCA contamination (Álvarez-Rodríguez et al., 2002; Cappellin et al., 2020; Salvatella et al., 2019). Several fungal genera, including *Paecilomyces*, *Penicillium*, *Mucor*, and *Trichoderma* have been reported to include species that can catalyze the O-methylation of TCP to TCA, with certain *Trichoderma* and *Fusarium* strains exhibiting particularly high bioconversion efficiencies. In cork oak ecosystems, diverse fungal communities have been documented on raw cork; however, detailed knowledge of their distribution within standing trees and across different forest sites remains limited (Serra et al., 2008; Goto et al., 2023; Veloso et al., 2024; Ruiz-Muñoz et al., 2025). Forest management strategies, such as grazing and selective clearing, can directly influence

the understory vegetation and indirectly shift microclimatic conditions thus affecting the diversity and activity of microflora associated with raw cork (Serra et al., 2008; Wan et al., 2022). Plant diversity in forests has been shown to affect both microbial diversity (Hiiesalu et al., 2017) and microbial function (Wu et al., 2019). Thus, the influence of the understory vegetation on microbial diversity in cork oaks and its consequent effect on TCA production is an important area of research that could provide valuable insights for the winemaking industry.

A common belief among cork processors and traders is that cork from forests with understory is more contaminated with TCA than cork from forests without understory (Cappellin et al., 2020). Whether variation in microbial communities driven by forest management practices may be responsible for TCA contamination has yet to be experimentally demonstrated.

This study aims to characterize the fungal diversity in cork collected from standing trees in managed cork oak forests, both with and without understory vegetation, using classical microbiological techniques to isolate and identify fungal taxa, and to assess differences in TCA contamination between two forest types. Additionally, it seeks to evaluate how fungal communities in the raw cork of standing trees may change based on forest management practices, position on the stem, and along the cork plank.

## 2. Materials and methods

### 2.1. Study sites

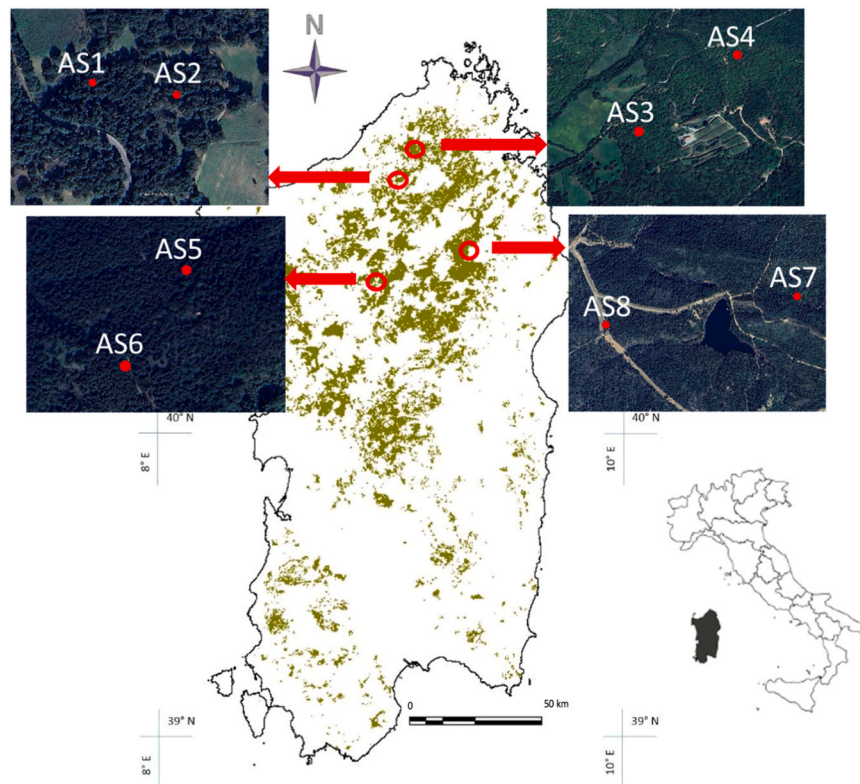
The survey was carried out in a total of eight cork oak stands in northern Sardinia (Italy), including four with understory and four without shrubs (Fig. 1). The sites were arranged in pairs: each site in a pair was in the same contiguous oak woodland stand and differed from the other site in the pair only in the presence or absence of the understory. This approach allowed us to minimize environmental variability among sites subjected to different understory management (Table 1).

Cork oak areas with understory comprised a dominant layer of *Quercus suber* with some *Quercus ilex* and *Quercus pubescens* trees intermixed among the cork oaks. The shrub layer consisted mainly of Mediterranean plant species typical of acidophilic substrates, including *Erica arborea*, *Arbutus unedo*, *Phillyrea angustifolia*, *Cytisus villosus*, *Cistus monspeliensis*, *C. incanus*, *Crataegus monogyna*, *Smilax aspera*, and *Rubus ulmifolius* (Table 1). Cork oak woodlands without understory were located in areas managed as silvicultural systems that integrate extensive livestock grazing typically by cattle with economically relevant cork production. Tree density ranged from 175 to 509 trees per hectare, corresponding to an estimated canopy cover of approximately 80–90 % (Table 1). In plots with lower tree density, individual crowns had a larger area, effectively occupying the canopy layer. The understory in the study sites has been persistently absent for at least the past ten years. Where grazing did not fully suppress shrub regrowth annually, any emerging understory vegetation was periodically removed through mechanical clearing, ensuring the long-term maintenance of the “no-understory” condition. (Table 1). In the two management types considered, tree regeneration occurred both sexually (from seed) and asexually (from sprouts on coppices).

The geological and pedological substrate was the same for all sites, composed mainly by intermediate-acid plutonic rocks, with a moderately coarse soil texture and granulometric class varying from loam to sandy loam (Table 1). Surface stoniness varied up to 40 %, with no outcropping rock (Madrau et al., 2006).

### 2.2. Cork sampling

In each area, ten cork oak trees were selected based on diameter at breast height (DBH), for a total of 80 trees, choosing individuals with representative and similar diameters within a maximum variation of 5 cm, and the absence of visible defects on the bark, and age of the cork



**Fig. 1.** Geographic distribution of the study sites in Sardinia. Each forest area is represented by paired plots with and without shrubby understory vegetation: *Padulo* (AS1, understory; AS2, no understory), *Baldu* (AS3, understory; AS4, no understory), *Fiorentini* (AS5, understory; AS6, no understory), and *Coiluna* (AS7, understory; AS8, no understory). The green area indicates the distribution of *Quercus suber* in Sardinia.

layer, ranging between 11 and 13 years. Four cork samples, measuring approximately 15 cm in height and 20 cm in width, were taken from each tree. Two samples were used for the analysis of fungal communities, one for the measurement of TCA, and one for assessing the presence and intensity of TCA-related sensory perception, which was evaluated by a panel of six expert judges through direct olfactory analysis. Cork samples used for the isolation of fungal communities were collected from both the lower (i.e. excluding the raw peduncle) and the upper part of the cork planks on the trunk. Samples used for the assessment of TCA were collected from the bark area between the two sampling points previously described (Fig. 2).

### 2.3. Sampling procedure

A standardized sampling protocol was implemented to guarantee methodological consistency, minimize environmental variability, and prevent cross-contamination, as described below and illustrated in Fig. 2:

- Samples were collected from the North-facing side of the tree to target consistently cooler, more humid microhabitats and thus minimize environmental variability among samples.
- The axe blade used for sampling was sterilized with hydrogen peroxide solution (36 volumes) using filter paper. Sterilization was performed at the beginning of the sampling process and repeated for every sample collection (both from the same tree and between different trees).
- Operators wore sterile gloves throughout the sampling process to prevent contaminations and changed gloves between each sample collection.
- Samples were minimally handled and immediately placed into appropriately labelled and sealed disposable plastic bags for

transport to the laboratory inside a cooler. For fungal biodiversity analysis, samples were processed on the same day of collection, within a maximum of 24 h from sampling. The sampling sites were visited on different days to ensure that all samples could be processed fresh, minimizing changes in fungal communities. To avoid increasing moisture and cross-contamination in the laboratory, samples were opened only when necessary and kept separated from each other throughout processing.

### 2.4. Fungal isolation

Fungal isolations from raw cork samples were carried out under sterile conditions using a vertical laminar flow hood. Small fragments of approximately 3 mm<sup>2</sup> were carefully cut from cork samples and placed on 90 mm Petri dishes containing Rose Bengal Agar with chloramphenicol (RBA + antibiotics) at pH 7.2 (Ruiz-Muñoz et al., 2025). A total of 960 Petri dishes were used for fungal isolation, including six plates per sample, each containing eight cork fragments. This resulted in 48 cork fragments per sample, consisting of 16 fragments from the outer bark, 16 from the core, and 16 from the middle section of the cross-section. In total 7680 cork fragments were processed for isolation of the fungal community.

Each Petri dish was labelled with unique IDs identifying the sample, the section of origin, the tree, and the sampling point (upper or lower part). The plates were then incubated in a climate-controlled chamber at 25°C and monitored daily to observe fungal growth (Prak et al., 2007; Maggi et al., 2008). Within ten days, the fungal colonies that developed were subcultured by transferring a small piece of agar with mycelium from the edge of the colony onto new 90 mm Petri dishes containing Potato Dextrose Agar (PDA), incubated at 25°C (Alvarez-Rodriguez et al., 2002). Pure cultures grown on PDA were transferred on the same day to ensure synchronous colony development. After 4–7 days,

**Table 1**  
Description of Sardinian cork oak sites investigated in this study.

| Site                            | Padulo (AS1)                        | Padulo (AS2)                        | Baldu (AS3)                         | Baldu (AS4)                         | Fiorentini (AS5)                  | Fiorentini (AS6)                  | Coiluna (AS7)                     | Coiluna (AS8)                     |
|---------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|-----------------------------------|-----------------------------------|-----------------------------------|-----------------------------------|
| GPS coordinates                 | N 40°56'44.37''<br>E 009°05'34.37'' | N 40°56'43.78''<br>E 009°05'38.36'' | N 40°51'20.99''<br>E 009°04'25.15'' | N 40°51'29.55''<br>E 009°04'40.41'' | N 40°30'50.3''<br>E 009°02'32.5'' | N 40°30'43.8''<br>E 009°02'28.2'' | N 40°36'45.9''<br>E 009°20'55.4'' | N 40°36'41.8''<br>E 009°20'18.7'' |
| Understory                      | Shrub                               | Not shrub                           | Shrub                               | Not shrub                           | Shrub                             | Not shrub                         | Shrub                             | Not shrub                         |
| Shrub species                   | a                                   |                                     | b                                   |                                     | c                                 |                                   | d                                 |                                   |
| Shrub canopy cover (%)          | 90                                  | 0                                   | 95                                  | 0                                   | 95                                | 0                                 | 95                                | 0                                 |
| Healthy status                  | Declining                           | Declining                           | Declining                           | Declining                           | Declining                         | Healthy                           | Healthy                           | Healthy                           |
| Altitude m (a.s.l.)             | 418                                 | 416                                 | 320                                 | 323                                 | 754                               | 763                               | 691                               | 671                               |
| Exposure                        | East                                | Southeast                           | Southeast                           | West                                | South                             | North-East                        | North-East                        | West                              |
| Slope                           | 5 %                                 | 3 %                                 | 3 %                                 | 5 %                                 | 8 %                               | 10 %                              | 10 %                              | 5 %                               |
| Substrate                       | Granitic                            | Granitic                            | Granitic                            | Granitic                            | Granitic                          | Granitic                          | Granitic                          | Granitic                          |
| Shrub control                   | No treatments                       | Livestock/Cut/Soil Tillage          | No treatments                       | Livestock/Cut                       | No treatments                     | Livestock/Cut                     | No treatments                     | Livestock                         |
| Grazer                          |                                     | Cattle                              |                                     | Cattle                              |                                   | Cattle                            |                                   | Cattle                            |
| Grazing intensity               |                                     | ≈ 0.5 LU/Ha, seasonal               |                                     | < 0.5 LU/Ha, seasonal               |                                   | < 0.5 LU/Ha, seasonal             |                                   | < 0.5 LU/Ha, seasonal             |
| Forest type                     | High forest                         | Irregular high forest               | Irregular high forest               | High forest                         | Irregular high forest             | High forest                       | High forest                       | High forest                       |
| Cork oak trees per hectare      | 509                                 | 382                                 | 493                                 | 294                                 | 310                               | 1512                              | 414                               | 175                               |
| Mean DBH                        | 28                                  | 31                                  | 22.3                                | 30.4                                | 27.6                              | 23.5                              | 19.5                              | 28.3                              |
| Basal area (m <sup>2</sup> /ha) | 31.3                                | 28.8                                | 19.2                                | 21.3                                | 18.5                              | 65.6                              | 12.4                              | 11                                |
| Mean height (m)                 | 8.5                                 | 7.1                                 | 7                                   | 9.2                                 | 7.1                               | 7.2                               | 5.8                               | 6.8                               |
| Cork age (Years)                | 12                                  | 12                                  | 12                                  | 12                                  | 11                                | 11                                | 13                                | 13                                |
| Annual mean temperature         | 12.9 °C                             | 12.9 °C                             | 14.8 °C                             | 14.8 °C                             | 14.5 °C                           | 14.5 °C                           | 13 °C                             | 13 °C                             |
| Annual rainfall                 | 723 mm                              | 723 mm                              | 716 mm                              | 716 mm                              | 1049 mm                           | 1049 mm                           | 1079 mm                           | 1079 mm                           |

a. *Cytisus villosus* Pourr, *Erica arborea* L., *Cistus monspeliensis* L., *Crataegus monogyna* Jacq., *Rubus ulmifolius* Schott., *Smilax aspera* L.

b. *Arbutus unedo* L., *Erica arborea* L., *Cytisus villosus* Pourr, *Cistus monspeliensis* L., *Crataegus monogyna* Jacq., *Phillyrea latifolia* L., *Rubus ulmifolius* Schott., *Smilax aspera* L.

c. *Erica arborea* L., *Arbutus unedo* L., *Cytisus villosus* Pourr, *Cistus monspeliensis* L., *Crataegus monogyna* Jacq., *Rosa canina* L., *Phillyrea latifolia* L., *Prunus spinosa* L., *Rubus ulmifolius* Schott., *Smilax aspera* L.

d. *Erica arborea* L., *Arbutus unedo* L., *Cytisus villosus* Pourr., *Cistus monspeliensis* L., *Daphne gnidium* L., *Crataegus monogyna* Jacq., *Helichrysum italicum* Roth., *Phillyrea angustifolia* L., *Rubus ulmifolius* Schott., *Smilax aspera* L.





**Fig. 2.** Cork sampling from selected trees: (A) field removal of the outer bark layer using an axe; (B) freshly harvested cork sample is placed in a disposable bag to ensure proper preservation until laboratory processing; (C) the three sampling points along the tree trunk; (D) cross-section of the cork showing the layers used for fungal community analysis and TCA detection.

subcultures were grouped into different morphotypes based on cultural characteristics (i.e. colony colour and texture, border type, and radial growth rate) and micromorphological features. Representative isolates of each morphotype, defined by these morphological traits, were selected for further molecular analyses.

## 2.5. DNA extraction, PCR amplification and sequencing

Genetic identification of fungal isolates was performed by PCR amplification, sequencing, and analysis of the internal transcribed spacer (ITS1 and ITS2) regions and the 5.8S rDNA gene. The DNA was extracted from 7-day old pure cultures grown on PDA using the InstaGene Matrix kit (BioRad Laboratories, Hercules, CA), following the manufacturer's instructions. Extracted DNA was stored at  $-20^{\circ}\text{C}$ .

The ITS regions were amplified by polymerase chain reaction (PCR) using the universal primers ITS1 and ITS4 (White et al., 1990). The reaction was conducted in a total volume of 50  $\mu\text{l}$  containing 18.2  $\mu\text{l}$  of  $\text{H}_2\text{O}$ , 10  $\mu\text{l}$  of 5x buffer, 5  $\mu\text{l}$  of dNTPs, 5  $\mu\text{l}$  of BSA, 5  $\mu\text{l}$  of each primer, 0.3  $\mu\text{l}$  of GoTaq polymerase (Promega), and 1.5  $\mu\text{l}$  of DNA template. The amplification was performed in a thermocycler (Hybaid PCR Express) using the thermal cycling conditions described by Fan et al. Aliquots of 5  $\mu\text{l}$  of each amplification product were visualized with gel electrophoresis with the 1 kb Plus DNA Ladder (InvitrogenTM) on a 1.5 % agarose gel in TAE buffer (40 mM Tris-acetate, 1 mM EDTA, pH 8.3) with SYBR-Safe (InvitrogenTM) as the molecular tracer. The amplicon size and concentration were determined using the Molecular Imager® Gel DocTM XR+ system with Image LabTM software (BIO-RAD).

PCR products were purified using the EUROGOLD Gel Extraction Kit (EuroClone) according to the manufacturer's instructions. Purified products were quantified and sent to BMR Genomics s.r.l. ([www.bmr-genomics.it](http://www.bmr-genomics.it)) for Sanger sequencing. Both forward and reverse sequences were read and analysed using BioEdit v. 5.0.6 (Hall, 2001). The resulting nucleotide sequences were compared with available data in the GenBank database using the BLAST algorithm (<http://blast.ncbi.nlm.nih.gov>) and the UNITE database (v10.0) (Abarenkov et al., 2024).

A threshold  $\geq 99\%$  was adopted for sequence validation, considering taxonomic annotation assigned to each query at both genus and species level. However, species-level identification was only accepted when no other species showed comparable homology at this threshold. For sequence identities between 98 % and 96 %, only the genus was accepted. For sequence similarity below 96 %, isolates were identified at order or family level or as “unknown” fungi (Ruiz-Muñoz et al., 2025).

All sequences were deposited in GenBank (<http://www.ncbi.nlm.nih.gov/>), with accession numbers (PQ770961 - PQ771108). Representative isolates are stored in the culture collection of the Servizio della Ricerca per la Sughericoltura e la Selvicoltura, Agris Sardegna.

## 2.6. TCA analysis

After the collection, cork samples were transferred to the laboratory and were open-air dried in a ventilated space, with the outer crust facing

upward, for 5–7 days until reaching a moisture content of approximately 10 %. Moisture levels were monitored daily using an Aqua Boy electronic hygrometer (5–20 % measurement range). Once the desired moisture content was achieved, the cork samples were stored on metal shelves in a closed, aerated environment until sample preparation for laboratory analysis began.

Samples preparation involved cutting approximately 200 g of cork from each sample, across its entire thickness. Using a circular saw, cork samples were cut into strips about 40 mm wide, followed by a cut using a band saw to obtain pieces 10–15 mm thick. These pieces were then ground with a Retsch SM 100 knife mill fitted with a 4 mm mesh sieve. To prevent cross-contamination, the grinding chamber, sieve, and collection container were thoroughly cleaned after each sample, and the initial granules produced were discarded. The resulting cork granules were stored in individual aluminium containers and sealed in 3 M Steri-Dual bags until analysis.

Headspace solid-phase micro extraction (HS-SPME) coupled to gas chromatography-single quadrupole mass spectrometer (GC/MS) has been used for TCA determination (Boutou and Chatonnet, 2007; Moreira et al., 2016; Tarasov et al., 2017) using a Perkin Elmer Clarus 690 gas chromatograph, equipped with a Clarus SQ 8 S mass spectrometer and a Gerstel MPS 2 auto sampler. TCA was extracted using a 100  $\mu\text{m}$  polydimethylsiloxane (PDMS) SPME fiber (Supelco) (Bianco et al., 2009). Analytes separation was performed on a Perkin Elmer Elite-5MS column (30 m length, 0.25 mm i.d., 0.25  $\mu\text{m}$  film thickness).

The analytical procedure was optimized maximizing instrument response through chemometric analysis, and validated determining key parameters: detection and quantification limits, linearity, precision, accuracy, recovery percentage, selectivity, robustness, and measurement uncertainty (De Aguiar et al., 1995; Leardi, 2009; Hibbert, 2012; Narendran et al., 2019; Lee et al., 2022). Factors affecting TCA extraction and desorption efficiency from the HS-SPME fiber were optimized using D-Optimal Experimental Design, involving variations in NaCl content (0–2.8 g), incubation time (0–20 min), adsorption temperature (30–60  $^{\circ}\text{C}$ ), adsorption time (15–60 min), agitation speed (0–700 rpm) and desorption time (20–300 s at 250  $^{\circ}\text{C}$ ).

For instrumental analysis, 5 g of cork granules were placed in glass jars filled with absolute ethanol and sealed with metal caps lined with Teflon discs to prevent TCA absorption. The samples were incubated at room temperature with agitation for  $24 \pm 2$  h. The extracts were diluted with water to adjust ethanol concentration to 12 % (V/V) 8 mL of cork extract and 100  $\mu\text{l}$  of internal standard (2,3,6-TCA) were placed in a 20 mL vial containing 2.8 g of NaCl. To ensure comparability of results with other laboratories and to confirm the performance of the method in terms of accuracy, an interlaboratory test consisting of 40 experiments was carried out in collaboration with an in-house laboratory of a cork factory.

## 2.7. Sensory analysis

All samples collected were individually transferred into 100-micron

polypropylene (PP) bags and vacuum-sealed (99.9 %). Subsequently, the samples were frozen and stored at  $-20^{\circ}\text{C}$  until their preparation for analysis (Vilia et al., 2024). All sensory tests were conducted in a test room specifically designed for sensory analysis ISO 8589:2007, equipped with individual tasting booths where temperature and humidity were controlled.

Upon thawing, the samples were measured and cut into six sub-samples of approximately  $5 \times 6$  cm. During pre-preparation, knife and cutting instruments used between samples were sanitized by wiping them with denatured alcohol before and after each cutting to prevent cross-contamination. The whole sub-samples surface was subjected to ultraviolet (UV) treatment and subsequently stored in hermetically sealed, coded food containers and kept in a climate-controlled chamber at  $20^{\circ}\text{C}$  until pre-analysis conditioning (Vilia et al., 2024).

Before the sensory analysis, samples were conditioned in a refrigerated thermostat at  $25.0 \pm 0.5^{\circ}\text{C}$  for  $24 \pm 2$  h and then maintained at room temperature for 30 min post-thermostating (Vilia et al., 2024). The panel consisted of six expert judges (four women and two men) trained ISO 8586:2023, specifically to identify off-flavors attributable to mould in cork/wine matrices, including TCA, methyl isoborneol (MIB), geosmin (GEO), 1-octen-3-ol, 1-octen-3-one, and pyrazines ISO (2230)8–1: (2021).

Each judge analyzed each sub-sample, identifying the presence or absence of TCA through direct olfactory sensory analysis and rating the perceived intensity on a five-point scale (1 = trace; 2 = slight; 3 = moderate; 4 = strong; 5 = very strong) ISO 4121:2003. The analysis was performed on all portions of the sample (six sub-samples), including the interior, sides, and crust. The presence of TCA was confirmed if at least one sub-sample tested positive for the judges, and the overall sample intensity was determined by the average intensity of the positive sub-samples.

## 2.8. Statistical analysis

We evaluated the effect of three factors on cork-associated fungal communities, management model (presence vs. absence of understory), sampling point (upper vs. lower section of the cork plank), and section of the cork sample (outer bark, middle section, and core), using measures of OTU (Operational Taxonomic Units, i.e., clusters of closely related DNA barcode sequences used as proxies for taxonomic groups at the species, genus, or higher levels), richness, trophic-groups compositions and abundance, and three proxies for cork taint (quantified TCA, sensory TCA, and TCA presence/absence with 0 indicating absence and 1 indicating presence). All statistical analyses were carried out in RStudio (V4.3.0) (R Core Team, 2023), with the Tidyverse (V2.0.0) (Wickham et al., 2019), vegan (V2.6–6) (Oksanen et al., 2024), and indicspecies (V1.7.14) (Cáceres and Legendre, 2009) packages. Shapiro-Wilk test was applied to assess normality of numerical variables before conducting further statistical tests. A one-way ANOVA test was conducted to understand if variation in TCA levels was influenced by the management of cork oak woodlands. This statistical test was performed three times, first using the level of TCA collected from the TCA analyses, second using the level of TCA recorded by the sensory analyses, and third using presence/absence of TCA.

Alpha diversity estimation was conducted using Shannon, Chao1, and Pielou indices (Ruiz-Muñoz et al., 2025). To understand how the composition of the microbial community influenced levels of TCA, a non-metric multidimensional scaling (NMDS) was performed. A permutational multivariate analysis of variance (PERMANOVA) was then applied to test if statistically significant differences were occurring in the observed beta-diversity. As we knew a-priori that the data was stratified according to the selected independent variables, we included the “strata” parameter in the model as a fixed variable (McArdle and Anderson, 2001). Beta diversity was evaluated using the Lennon, Sørensen, and Jaccard indices, and ANOVA was performed to test for correlation with management models, sampling points and cork sample

sections groups. For those combinations in which we observed statistically significant differences, we further employed the Tukey’s HSD. We assessed variation in fungal-community OTU composition and explored its relationship with cork taint (TCA) with similar approach by combining multivariate ordination, hypothesis testing, indicator-species analysis, and functional annotation. NMDS on a Hellinger-transformed OTU abundance matrix, fitting scaled environmental variables, including management model, sampling point, and cork section, by correlating NMDS axis scores with explanatory factors via ANOVA. We tested for significant differences in microbial community composition among treatment groups using PERMANOVA.

To identify taxa most strongly associated with each factor, we carried out an indicator-species analysis (ISA), then assigned ecological guilds to those taxa using FUNGuild v1.2 (Nguyen et al., 2016).

In addition, we evaluated how fungal-community composition shapes the distribution of sensory-detected TCA by constructing two separate datasets. The first dataset, we pooled OTU abundance across all three sections per tree, thereby capturing the combined influence of shrub-layer management and sampling point, by summing counts from the outer bark, middle section and core. In the second dataset, we retained sample section level (i.e. outer bark, core and middle section of the cross-section) to assess how within-cork spatial variability influences TCA presence and intensity.

## 3. Results

### 3.1. Fungal isolation and community diversity

A total of 7370 of fungal isolates were obtained from raw cork samples, 47.3 % of which came from cork oak areas without understory and 52.7 % from cork oak stands with understory vegetation. Location of the sample on the cork section (i.e. outer bark, middle section and core) had the greatest effect on isolation success: 53 % of the isolations were obtained from the outer bark, 25.8 % from the middle section and 21.2 % from the core. Sampling height did not influence isolation percentage and 50.8 % of isolates came from samples collected lower on the trunk, while 49.2 % of isolates came from samples collected higher on the trunk. Molecular analyses of ITS sequences identified 151 OTUs, including 37 different families and 49 genera. Genetic identification at the species level was achieved for 91 OTUs. Ascomycota (81.6 %) was the predominant phylum, followed by Mucoromycota (10.5 %), Basidiomycota (5.3 %) and Mortierellomycota (2.6 %).

The Chao1, and Shannon indices (157.56 and 3.24, respectively) confirmed the consistent abundance and diversity of fungi across the investigated sites. Additionally, the Pielou (0.65) index highlighted a well-balanced distribution of OTUs within the community. Notably, the Baldu site exhibited the highest fungal abundance. While species richness was generally higher in study areas without understory, fungal abundance was greater where understory was present, except for the AS3 area (Supplementary material 1). The most frequently isolated OTUs in terms of both frequency and abundance were: *Penicillium glabrum*, *P. angulare*, *Mucor plumbeus*, *M. fragilis*, *Mucor* sp.1, *Trichoderma simmonsii*, *T. paraviridescens*, and *Cladosporium cladosporioides* (Supplementary material 6).

### 3.2. Functional groups of the fungal community

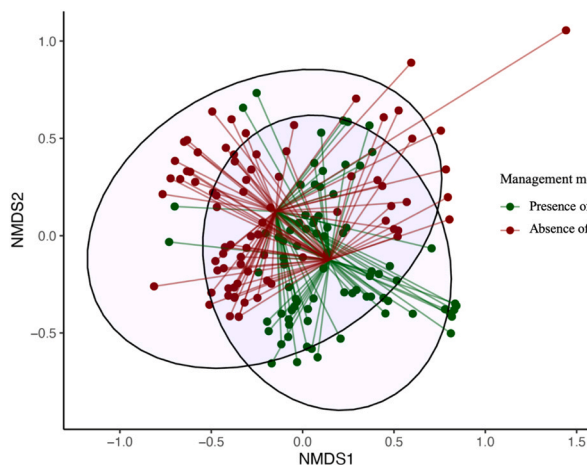
The functional annotation of the isolated fungi was performed using the FUNGuild database, which classified fungi according to their ecological role and trophic mode (Supplementary material 7). The classifications included: soil saprotrophs (1.3 %), litter saprotrophs (1.3 %), lichen parasites (5.3 %), epiphytes (8.6 %), animal parasite (13.9 %), animal pathogen (16.6 %), plant saprotrophs (17.2 %), dung saprotrophs (17.9 %), fungal parasites (23.8 %), wood saprotrophs (32.5 %), undefined saprotrophs (33.1 %), endophytes (43 %), and plant pathogens (48.3 %). Supplementary material 8 illustrates the

proportion of ecological roles assigned to the fungal species isolated from the cleared cork oak (M1) and scrubland habitats (M2). The proportion of ecological roles remains largely unchanged between the two different management models. The functional annotation performed on the fungal community sampled from the cork section isolation points (Supplementary material 9) revealed a greater diversity of ecological role patterns in the core and mid-section communities compared to the one obtained from the outer bark. Fungal taxa commonly associated with latent pathogenicity, such as species from the genera *Diplodia*, *Biscogniauxia*, *Botryosphaeria*, *Dothiorella* and *Discula*, were isolated across multiple sites and tree sections (Supplementary material 2), showing a variable distribution pattern.

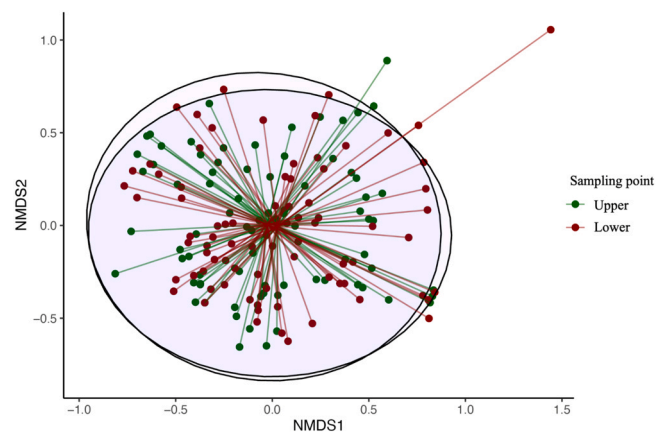
### 3.3. Fungal community variation in cork across oak forests and across sampling locations on trees

There were no significant differences in OTU richness between the cork oak forests with and without understory vegetation. However, analysis of fungal communities in relation to management type, sampling height, TCA values, and sampling points yielded several key insights. NMDS analysis indicated a significant influence of management type on OTU composition in fungal communities ( $p$ -value=0.066), with variations attributable to the presence or absence of understory (Fig. 3). In contrast, sampling height did not significantly impact fungal community composition ( $p$ -value = 0.99) (Fig. 4). These findings were confirmed by PERMANOVA, which confirmed the significant effect of management type on OTU composition (Table 3), while sampling height had no significant influence (Table 3).

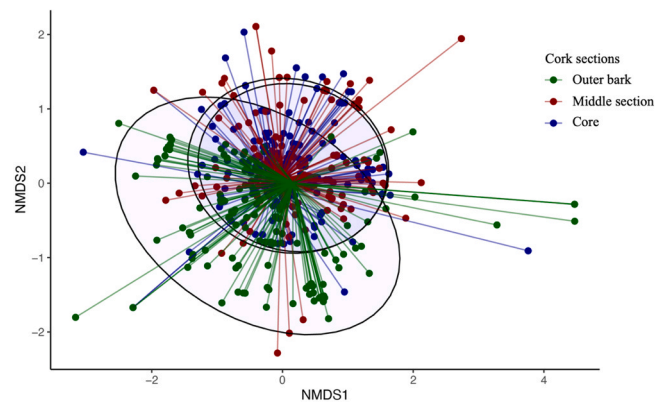
Sample location in the cork section (core, middle section and outer bark) affected OTU composition in associated fungal communities. Species richness ( $S_r$ ) increased when moving from the outermost to the innermost section as evidenced by the fact that 83 OTUs were isolated from the outer bark section, 95 from the middle section, and 99 from the core section. Shannon ( $H'$ ) and Pielou ( $J$ ) indices revealed that, despite having fewer OTUs, the outer bark community was better balanced and exhibited higher biodiversity compared to the core and middle section communities (Supplementary material 1). The NMDS (Fig. 5) and PERMANOVA (Table 3) confirmed that fungal community composition differed significantly between the outer bark and both the middle ( $p$ -value = 0.003) and core ( $p$ -value = 0.003) sections, with a less pronounced but still significant difference between the middle and core ( $p$ -value = 0.027).



**Fig. 3.** Non-metric multidimensional scaling (NMDS) ordination comparing the microbial community composition under two management regimes: presence of understory (green) and absence of understory (red). Thin lines connect each sample to its group centroid, while shaded ellipses represent 95 % confidence intervals around the centroids for each management type.



**Fig. 4.** NMDS ordination of the sampling. Points represent individual samples collected from the upper (green) and lower (red) parts of the tree relative to the debarking height. NMDS1 and NMDS2 axes summarize differences in community composition. Thin lines connect each sample to its group centroid, while shaded ellipses represent 95 % confidence intervals around the centroids for each height level.



**Fig. 5.** Sampling point ordination using non-metric multidimensional scaling (NMDS). Each point is color-coded according to the cork section: outer bark (green), middle section (red), and core (blue). The two axes (NMDS1 and NMDS2) summarize variation in community composition among samples. Thin lines connect each sample to its group centroid, while shaded ellipses represent 95 % confidence intervals around each centroid.

### 3.4. Fungal community composition, but not its diversity, is associated with varying TCA levels

Chemical analyses revealed that the OTUs fungal community composition in the two-forest management systems were associated with different TCA levels per tree within each system (2). While ANOVA assessed group homogeneity in relation to management, PERMANOVA tested differences in community composition between groups. Non-significant PERMANOVA results (Table 3) indicate the overall OTUs composition does not differ based on presence or intensity of TCA off-odours. Instead, significant changes are related to variations in the abundance and frequency of OTUs within the fungal community.

### 3.5. Effect of understory vegetation on TCA levels

One-way ANOVA showed that the level of TCA was influenced by the management of the understory vegetation. When comparing the level of TCA recorded in areas with and without understory, the levels of TCA were significantly different. The ANOVAs identified significant difference between the two cork oak forest types when analysing the



numerical TCA (p-value < 0.05), recorded by both the sensorial analysis (p-value < 0.05) and the presence/absence of the TCA molecule (p-value < 0.05) (Table 2). TCA levels were higher in sites with understory, suggesting that removing such vegetation might have a beneficial outcome by reducing the level of TCA molecule in the harvested cork. Notably, TCA values were higher in cork samples from areas with understory: the mean value per plant in this management type was 90.84 ng/L, compared to 7.18 ng/L in areas without understory. Furthermore, the ANOVA test confirmed significant differences in TCA levels per individual tree (Table 2) and in the presence and intensity of TCA off-odours (Table 2) between the two management types, highlighting the impact of understory presence on TCA levels.

### 3.6. Fungal indicator species across forest management and cork sections

Indicator Species Analysis (ISA) (Cáceres and Legendre, 2009) identified OTUs associated with forest management type and sampling position on cork. The classical ISA revealed taxa characteristic of stands with or without understory vegetation. The results identified a total of 29 indicator species (IS), divided as follows between the two forest types: 14 OTUs within forest with understory and 15 OTUs in cleared cork oak forests (Supplementary material 3). ISA associated with each of the three positions on cork sections were evaluated using both the classical ISA (Supplementary material 4) and the combined group methods (Supplementary material 5). To gain further insights on the effect of position on cork section on IS, the second group of the combined method was set as the core and mid-section community, based on PERMANOVA's results (Table 3). The classical ISA identified a total of 32 indicator species: 24 from the outer bark, 3 from the middle section and 5 from the core. The combined groups ISA identified 39 IS. Group 1 was characterised by 29 IS: 25 OTUs from the outer bark, 1 from the mid-section and 3 from the core section; 10 IS were associated with the second group.

## 4. Discussion

This study provides valuable insights into the culturable fraction of the fungal community associated with cork oak, highlighting patterns potentially influenced by forest management and affecting cork quality. Our results indicate that forest management, i.e. the presence or absence of understory vegetation, shapes the composition and dynamics of the associated fungal community and may influences cork quality, under the specific environmental conditions considered, highlighting a critical issue at the base of the cork production chain. Fungal richness did not vary based on the understory, suggesting that the number of fungal species involved in TCA production may be too small to affect overall community richness. Fungal richness was comparable across various oak stands, consistent with previous studies showing that fungal

**Table 2**  
Results of ANOVA analyses testing the effects of understory management and sensory off-odor features on TCA concentrations.

| Analysis <sup>a</sup> | Independent Variable     | Variable          | R2    | F      | p-value <sup>b</sup> |
|-----------------------|--------------------------|-------------------|-------|--------|----------------------|
| A                     | Management               | TCA concentration | 0.066 | 8.272  | <b>0.005</b>         |
| B                     | Off-odor presence (TCA)  | TCA concentration | 0.171 | 16.051 | <b>0.000</b>         |
| C                     | Off-odor intensity (TCA) | TCA concentration | 0.150 | 13.738 | <b>0.000</b>         |

<sup>a</sup> A: evaluates differences in TCA concentration between plots with and without understory vegetation; B and C: assess the relationship between TCA concentrations and the presence or intensity of perceived off-odors, respectively.

<sup>b</sup> Bold p-values indicate statistically significant effects (p < 0.05), meaning that TCA concentrations differed significantly between the compared groups (e.g., management with vs. without understory vegetation).

**Table 3**  
Results of PERMANOVA analysis.

| Analysis <sup>a</sup> | Variable                | R2      | F      | p-value      |
|-----------------------|-------------------------|---------|--------|--------------|
| A                     | Management              | 0.04182 | 6.8963 | <b>0.001</b> |
| B                     | Sampling height         | 0.00327 | 0.5176 | 0.994        |
| C                     | Management              | 0.0504  | 6.2948 | <b>0.001</b> |
|                       | TCA                     | 0.01273 | 1.5903 | <b>0.022</b> |
| D                     | TCA off-odors presence  | 0.01748 | 1.3881 | 0.097        |
| E                     | TCA off-odors intensity | 0.01378 | 1.0899 | 0.33         |
| F                     | Cork section point      | 0.04544 | 10.877 | <b>0.001</b> |
| G                     | Core vs Middle          | 0.00904 | 2.7557 | <b>0.027</b> |
|                       | Core vs Outer bark      | 0.05841 | 18.981 | <b>0.003</b> |
|                       | Middle vs Outer bark    | 0.05963 | 19.405 | <b>0.003</b> |

The table reports the proportion of variance explained (R<sup>2</sup>), F-values, and associated p-values for each model. Analyses were performed to assess the influence of management practices (A, C), sampling height (B), TCA levels (C), off-odor presence and intensity (D, E), and cork section (F, G) on fungal community composition. Pairwise comparisons among cork sections (G) are also reported. Statistically significant effects (p < 0.05) are indicated in bold.

<sup>a</sup> Table 3: Results of PERMANOVA analyses testing the effects of multiple factors on fungal OTU community composition.

communities can respond dynamically to habitat modifications, ensuring their persistence across diverse environmental gradients (Tomao et al., 2020; Bahram and Netherway, 2022). However, indicator species analysis revealed clear differences between forests with and without understory: *M. plumbeus*, *M. fragilis*, *Fusarium* sp.2, and *Trichoderma paraviridescens* were characteristic of forests with understory, while *Mucor* sp.1 and *T. gamsii* were indicative of forests without understory. These findings indicate that, although overall richness remains stable, the composition of specific taxa is strongly influenced by management practices creating distinct ecological niches. The balanced distribution of OTUs, as indicated by the Pielou (J) index, further reflects the ability of fungal communities to partition resources and maintain equilibrium, based on their functional activities (Ruiz-Muñoz et al., 2025). The dominance of OTUs such as *Penicillium glabrum*, *P. angulare*, *Mucor plumbeus*, *M. fragilis*, *Mortierella* sp.1, *T. simmonsii*, *T. paraviridescens*, and *C. cladosporioides*, suggests that these species are particularly well-adapted to the specific ecological niches provided by the cork surface (Egidi et al., 2019; Bahram and Netherway, 2022).

The predominance of Ascomycota (81.6 %) within the fungal community is consistent with previous studies reporting fungal communities in cork oak forests (Ruiz-Muñoz et al., 2025), which are known to play essential roles in ecosystem processes (Egidi et al., 2019). The co-occurrence of Mucoromycota (10.5 %), Basidiomycota (5.3 %), and Mortierellomycota (2.6 %) highlights the taxonomic and potential functional diversity of the communities (Manici et al., 2024). The presence of these groups in the ecosystem and in the cork substrate can serve as indicators of potential functional roles, as they are known to contribute to key ecological processes, such as the decomposition of organic matter, including complex polymers such as lignin (Ruiz-Muñoz et al., 2025).

Functional annotation of the fungal community revealed a high representation of taxa commonly reported as latent pathogens. These organisms are often asymptomatic endophytes in healthy tissues but can express pathogenic traits under certain biotic or abiotic stress conditions (Franceschini et al., 2005; Moricca et al., 2016). Their consistent detection across all study sites reflects their ecological ubiquity in cork oak forests and highlights the importance of considering the dynamic behaviour of fungal communities in response to environmental pressures (Roberts et al., 2020). Besides latent pathogens, a large portion of the fungal community was composed of endophytes (43 %), which are known to support host tolerance to abiotic and biotic stresses (Terhonen et al., 2019), and saprotrophs (over 65 %), which contribute to organic matter decomposition and nutrient cycling, fundamental for forest productivity and soil health (Chen et al., 2022; Zhao et al., 2023). Other functional groups such as fungal parasites, dung saprotrophs, and



epiphytes reflected the ecological diversity of the fungal community and its potential role in ecosystem functioning (Shi et al., 2019). The wide range of functional groups detected reflects the complex ecological dynamics of cork oak-associated fungi and their involvement in key ecosystem processes. Analysis of fungal communities in relation to management type and sampling height, TCA level and sampling point yielded several additional key insights. NMDS and PERMANOVA analysis indicated a significant trend of effect of management type on OTU composition in fungal communities ( $p$ -value = 0.066), with variations attributable to the presence or absence of the understory. In contrast, sampling height did not significantly impact fungal community composition ( $p$ -value = 0.99). The lack of variation in fungal community composition along the height gradient contrasts with previous studies on the whole caulosphere or xylem, but may be explained by the unique structural and microclimatic conditions of the outer bark and by the good health status of all trees studied. Variation in fungal community composition and richness along the caulosphere has been reported for trees experiencing significant stem decay (Garbelotto and Johnson, 2023). Additionally, the thick cork layer provides a relatively homogeneous environment that likely buffers vertical differences along the stem, in contrast to the greater variability observed in wood tissues or epiphytic compartments. Previous studies have suggested that fungal communities in tree-associated habitats can be influenced by factors such as bark composition, exposure to sunlight, or chemical compounds (Hofmann et al., 2023), but these effects may be less pronounced in cork oak systems due to the specific physiological and structural traits of this species.

#### 4.1. Fungal community dynamics and TCA in raw cork

Based on the observed patterns in community composition and indicator species, we examined how shifts in fungal community composition may also be relevant for cork quality. In particular, the contribution of fungi to TCA contamination could depend on the relative prevalence of taxa groups with the metabolic capacity to produce this compound, potentially leading to positive or negative correlations between particular species and TCA levels in cork (Prak et al., 2007). The analysis of variance (ANOVA) revealed a significant association between TCA contamination in cork and the composition of fungal communities, suggesting that understory management may influence TCA production, either directly or indirectly through effects on microbial dynamics. Specifically, understory presence may create microhabitats with altered humidity, temperature (Tomao et al., 2020), and organic matter accumulation that favor fungal taxa capable of producing chloroperoxidases (CPOs), enzymes that catalyze the chlorination of phenols and lignin, forming 2,4,6-trichlorophenol (TCP), a direct precursor of TCA (Ruiz-Muñoz et al., 2025). This compound can subsequently be O-methylated by fungi or transformed through chemical reactions into TCA (Ruiz-Muñoz et al., 2025; Zhou et al., 2024). Notably, Ruiz-Muñoz et al. (2025) reported that over 70 % of fungal isolates from cork exhibited chloroperoxidase activity, underlining their potential contribution to TCA biosynthesis. Chemical data and NMDS analysis further supported the influence of shrub cover on fungal composition, with areas lacking understory showing distinct communities and reduced TCA levels (mean: 7.18 ng/L vs. 90.84 ng/L with understory). Although a direct correlation between overall fungal community composition and TCA intensity was not statistically significant, these results may reflect the complexity of interactions between specific taxa within the community and TCA dynamics which may be driven by a limited number of fungal species and not by an entire community (Barreto et al., 2011). The spatial distribution of fungal taxa across cork layers confirmed the role of substrate heterogeneity in shaping microbial communities. Differences between the outer bark and inner layers were attributed to gradients in moisture and nutrients, as observed in other forest systems (Chen et al., 2022; Gan et al., 2022; Hofmann et al., 2023). These patterns may help explain differential fungal activity and its impact on cork

properties, including potential off-flavor formation, as outer layers have been previously linked to higher TCA content (Rigaud et al., 1984).

ANOVA results revealed significant differences between forests with and without understory or all TCA measurements (numerical, sensory, and presence/absence;  $p < 0.005$ ), highlighting that understory management significantly influences TCA levels, a critical point at the base of the cork production chain. One-way ANOVA analyses confirmed that forests with unmanaged understory vegetation exhibit significantly higher TCA concentrations, both chemically and sensorially, than forests without understory. This finding indicates that the presence of low vegetation in cork oak stands could potentially create microhabitats that favor fungal activity associated with TCA biosynthesis, thereby amplifying the risk of contamination even before cork reaches industrial processing stages.

#### 4.2. Forest management strategies and TCA mitigation

TCA contamination is a primary determinant of cork stopper quality, which in turn drives the economic valuation of cork in the market (Gil, 2014; Veloso et al., 2024). Our results confirm the common knowledge in Mediterranean cork oak systems, where producers have consistently reported higher TCA incidence in unmanaged forests with low vegetation compared to regularly managed stands (Cappellin et al., 2020). These data provide quantitative confirmation of this widely acknowledged pattern, demonstrating that proactive understory management can serve as a strategic intervention to mitigate TCA accumulation at the forest level. Under the climatic conditions of the study area, understory management practices may reduce the favorable conditions for fungal taxa involved in chlorophenol transformation not only limiting initial TCA formation but also contributing to the production of higher-quality cork, ultimately enhancing the economic sustainability of the cork industry. These findings further reinforce the notion that understory management play a dual role: it reduces microbial drivers of TCA while preserving cork quality.

Mediterranean cork oak forests offer opportunities to integrate multifunctional practices, such as controlled grazing, which regulate shrub density, support biodiversity, and sustain local economies (Cappellin et al., 2020). Nevertheless, in sites where understory vegetation covers more than 90 % of the ground, above-ground shrub biomass may enhance TCA contamination by fostering favorable conditions for active fungal metabolism. While shrub roots and their ectomycorrhizal networks contribute positively to tree health and ecosystem resilience (Smith and Read, 2008; Reis et al., 2018; Sanz-Benito et al., 2023), their aerial parts require careful management.

While in the forests considered in this study, understory vegetation covering over 90 % of the ground may exacerbate TCA contamination, the root systems of understory vegetation play essential roles in ecosystem functioning. Through associations with ectomycorrhizal fungi, they enhance nutrient uptake, improve tree resilience to stress, support microbial diversity and organic matter turnover, and can favour cork oak natural regeneration and seedling survival (Smith and Read, 2008; Kennedy et al., 2012; Reis et al., 2018; Liang et al., 2020; Sanz-Benito et al., 2023). Therefore, forest management strategies should aim to balance the ecological benefits provided by the below-ground components of the shrub layer with the potential risks associated with excessive aboveground biomass. Adaptive silvicultural interventions, including targeted shrub control and integrated grazing, may reduce TCA risk while maintaining the multifunctionality and sustainability of cork oak ecosystems (Bugalho et al., 2011; Tarasov et al., 2022).

## 5. Conclusion

This study demonstrates that forest management practices shape fungal community composition, generating distinct ecological niches that may influence cork quality. Moving forward, further research is

needed to better understand the specific roles of key fungal species, particularly those involved in TCA production, and how their interactions are shaped by environmental variables such as soil moisture, temperature, tree canopy cover and silvicultural practices. Addressing these questions will require methodological approaches capable of distinguishing metabolically active fungi from dormant or inactive organisms. While DNA-based techniques provide a broad overview of fungal diversity, they cannot identify which taxa are functionally active. Integrating direct isolation with RNA-based transcriptomic approaches could better characterize biologically active fungal assemblages and their roles in processes such as TCA formation. Insights into these dynamics will be crucial for optimizing forest management strategies that enhance cork quality without compromising ecological balance. Integrating this knowledge into management models could significantly improve both the ecological integrity of cork oak forests and the economic sustainability of the cork industry. In this way, we can promote the long-term health of these ecosystems, fostering a balanced relationship between natural biodiversity and the needs of cork production.

### CRedit authorship contribution statement

**Salvatore Seddaiu:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Camilla Morittu:** Writing – review & editing, Writing – original draft, Validation, Supervision, Methodology, Formal analysis, Data curation, Conceptualization. **Franco Pampiro:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Margherita Addis:** Formal analysis, Data curation. **Marco Caredda:** Formal analysis, Data curation. **Francesca Meloni:** Writing – review & editing, Methodology, Formal analysis, Data curation. **Marco Serra:** Writing – review & editing, Methodology, Data curation, Conceptualization. **Matteo Garbelotto:** Writing – review & editing, Formal analysis, Data curation. **Bruno Scanu:** Writing – review & editing, Validation, Supervision, Methodology, Formal analysis, Data curation, Conceptualization. **Stefania Fanni:** Formal analysis, Data curation. **Vittorio Alessandro Marrone:** Formal analysis, Data curation. **Adriana Vilja:** Writing – review & editing, Validation, Methodology, Formal analysis, Data curation, Conceptualization. **Gianfranco Marzeddu:** Investigation, Data curation. **Luca Sarais:** Writing – review & editing, Methodology, Formal analysis, Data curation. **Antonio Mulas:** Formal analysis, Data curation. **Edoardo Scali:** Writing – review & editing, Methodology, Formal analysis, Data curation. **Giovanni Piras:** Writing – review & editing, Project administration. **Giuseppino Pira:** Investigation, Data curation. **Pino A. Ruiu:** Investigation, Data curation, Conceptualization.

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### Declaration of Competing Interest

The authors declare that they have no competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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### Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.foreco.2025.123350.

### Data availability

Data will be made available on request.

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